

Virgo SLM V1.0

Building a 120M Parameter Transformer Language Model from Scratch

A Self-Learning Engineering Project

Project Tagline: *"Not Everybody Needs The Blueprint."*

Report Theme

This report should feel like the engineering notebook of a first-year IIT Dharwad student. It should emphasize learning, experimentation, debugging, curiosity, and persistence rather than presenting the project as industrial research. Include screenshots, simple diagrams, handwritten-style flowcharts if possible, engineering notes, failed experiments, and lessons learned.

Project Roadmap (Full Page Diagram)

Virgo Development Phase I → Virgo Base → Language Learning ↓ Phase II → Virgo Chat → Chat Style Tuning ↓ Phase III → Virgo Instruct (IFT) → Instruction & Response Optimization

Phase I – Language Learning

Goal: Build the language foundation of Virgo. Training: • Approximately 2 Billion pretraining tokens
Topics to cover: • Dataset collection • Cleaning pipeline • Tokenizer training • Byte-Level BPE • Transformer implementation • Self-supervised pretraining • Grammar, vocabulary, syntax and semantic learning
Output: Virgo Base
Suggested diagrams: Dataset pipeline, tokenizer workflow, preprocessing flow, transformer overview.

Phase II – Chat Style Tuning

Goal: Transform Virgo Base into a conversational assistant. Training: • Approximately 230 Million conversational tokens
Topics to cover: • Chat datasets • Assistant identity • Multi-turn conversations • Writing improvements • Programming improvements • Conversation formatting • Better conversational flow
Output: Virgo Chat
Suggested diagrams: Conversation pipeline, chat dataset flow, SFT workflow.

Phase III – Instruction & Response Optimization

Goal: Teach Virgo to understand user intent and produce accurate instruction-following responses. Training: • Approximately 278 Million instruction-tuning tokens
Highlight: A large custom instruction dataset was designed and curated by the author through many months of manual work. Building, cleaning, expanding, and organizing this dataset required significant effort and several hours of engineering. This phase teaches Virgo to: • Follow user instructions • Produce structured outputs • Improve generalization • Better response formatting • Writing tasks • Coding tasks • Information extraction • Markdown generation • Exact-format responses
Output: Virgo Instruct (IFT)
Suggested diagrams: Instruction dataset pipeline, prompt→response workflow, dataset categories, training pipeline.

Recommended Chapters

- 1. Why I Started Virgo
- 2. Learning Journey
- 3. Three-Phase Development
- 4. Dataset Engineering

- 5. Tokenizer
- 6. Transformer Architecture
- 7. Training Journey
- 8. Challenges & Failed Experiments
- 9. Example Responses
- 10. Current Limitations
- 11. Future Vision
- 12. Personal Message

Current Limitations

Discuss honestly that Virgo currently does not support: • Web search • Image generation • File creation • Long-context conversations • Persistent memory • Continuous multi-session conversations • Advanced mathematical solving • High-level reasoning • Tool usage Mention that Virgo currently follows a prompt → response interaction pattern.

Future Vision

Discuss future improvements including: • Better reasoning • Larger context windows • Multimodal capability • Retrieval-Augmented Generation • Memory • Tool use • Better mathematics • Larger datasets • Scaling beyond the Small Language Model category while preserving efficiency.

Ending Message

End the report with a heartfelt personal note explaining that Virgo was developed independently as a self-learning project. Reconnect to the tagline: "Not Everybody Needs The Blueprint." Explain that every failed training run, preprocessing bug, CUDA error, and debugging session became part of the learning journey. Encourage other students to start building rather than waiting until they know everything. Let the final page leave readers with inspiration rather than technical details.